

Mechanical Engineering (Hinglish)

Basic Thermodynamics

DPP: 1

Basic Concept

- Q1** A rigid tank is divided into two compartments by a partition. One compartment contains 3 kmol of N_2 at 600 kPa and the other compartment contains 7 kmol of CO_2 at 200 kPa. Now the partition is removed, and the two gases form a homogenous mixture at 300 kPa. The partial pressure of N_2 in the mixture is
 (A) 75 kPa (B) 90 kPa
 (C) 150 kPa (D) 3175 kPa
- Q2** An 80 L rigid tank contains an ideal-gas mixture of 5g of N_2 and 5g of CO_2 at a specified pressure and temperature. If N_2 were separated from the mixture and stored at mixture temperature and pressure, its volume would be
 (A) 32 L (B) 36 L
 (C) 40 L (D) 49 L
- Q3** An ideal-gas mixture consists of 3 kg of Ar and 6 kg of CO_2 gases. The mixture is now heated at constant volume from 250 K to 350 K. The amount of heat transfer is
 (A) 374 kJ (B) 436 kJ
 (C) 488 kJ (D) 525 kJ
- Q4** An ideal-gas mixture consists of 30 percent Helium and 70 percent Argon gases by mass. The mixture is now expanded reversible adiabatically in a turbine from 400°C and 1.2 MPa to a pressure of 200 kPa. The mixture temperature at turbine exit is
 (A) 195°C (B) 56°C
 (C) 112°C (D) 130°C
- Q5** A piston-cylinder device contains an ideal-gas mixture of 3 kmol of He gas and 7 kmol of Ar gas at 50°C and 400 kPa. Now the gas expands at constant pressure until its volume double. The amount of heat transfer to the gas mixture is
 (A) 6.2 MJ (B) 4.2 MJ
 (C) 27 MJ (D) 67 MJ
- Q6** The temperature 't' on a thermometric scale is defined in terms of a property 'K' by the relation $t = a \ln K + b$ where 'a' and 'b' are constants. The values of 'K' are found to be 1.83 and 6.78 at the ice point and the steam point, the temperature of which are assigned the numbers 0 and 100 respectively. Determine the temperature corresponding to a reading of 'K' equal to 2.42 on the thermometer?___(in integer).
- Q7** A new scale N of temperature is divided in such a way that the freezing point of ice is 100°N and the boiling point is 400°N. What is the temperature reading on this new scale when the temperature is 150°C? At what temperature a both the Celsius and the new temperature scale reading would be the same?_____ (answer in integer).
- Q8** Match **List-I** with **List-II** and select the correct answer using the codes given below the lists.
- List-I (Parameter)**
- Volume
 - Density
 - Pressure
 - Work
- List -II (Property)**
- Path function
 - Intensive property
 - Extensive property
 - Point function
- (A) A-3, B-2, C-4, D-1



- (B) A-3, B-2, C-1, D-4
 (C) A-2, B-3, C-4, D-1
 (D) A-2, B-3, C-1, D-4

Q9 Consider the following statements:

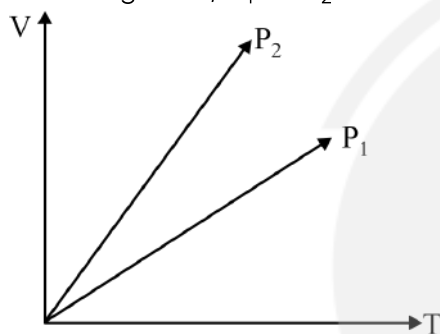
A real gas obeys perfect gas law at very

1. High temperatures
2. High pressures
3. Low pressures

Which of these statements is/are correct?

- (A) 1 alone (B) 1 and 3
 (C) 2 alone (D) 3 alone

Q10 The volume V vs. temperature ' T ' graphs for a certain amount of a perfect ideal gas at two pressures ' P_1 ' and ' P_2 ' are shown in the fig. Which is greater, ' P_1 ' or ' P_2 '



- (A) $P_2 < P_1$ (B) $P_2 > P_1$
 (C) $P_2 = P_1$ (D) No relation

Q11 Consider an ideal gas contained in vessel. If intermolecular interaction suddenly begins to act, which of the following happens?

- (A) The pressure increase
 (B) The pressure remains unchanged
 (C) The pressure decrease
 (D) The gas collapses

Q12 The correct sequence of the decreasing order of the value of characteristic gas constants of the given gases is

- (A) hydrogen, nitrogen, air, carbon dioxide
 (B) carbon dioxide, hydrogen, nitrogen, air
 (C) air, nitrogen, carbon dioxide, hydrogen
 (D) nitrogen, air, hydrogen, carbon dioxide

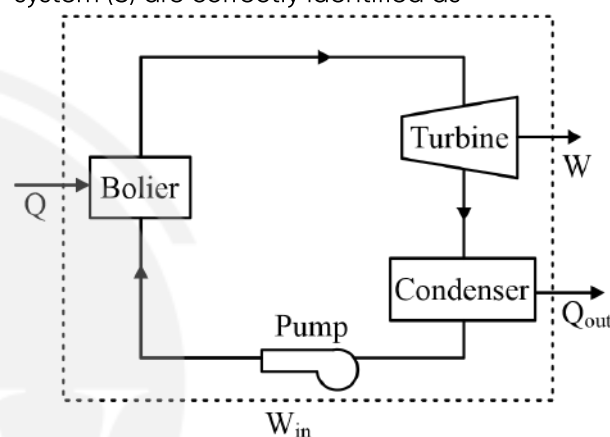
Q13 Consider the following conditions for the reversibility of a cycle:

1. The P and T of the working substance must not differ appreciably, from those of the surroundings at any state in the process.
2. All the processes, taking place in the cycle, must be extremely slow.
3. The working parts of the engine must be friction free.

Which of the above conditions are correct?

- (A) 1, 2 and 3 (B) 1, 2 and 3
 (C) 1 and 3 only (D) 2 and 3 only

Q14 In the figure showing a schematic of a thermal power plant, the closed system (S) and open system (S) are correctly identified as



- (A) boiler and condenser are open system and turbine, pump and overall plant are closed systems
 (B) turbine and pump are open systems and boiler, condenser and overall plant are closed system
 (C) boiler, turbine, condenser and pump are open systems and overall plant is a closed system
 (D) overall plant is an open system and boiler, turbine, condenser and pump are closed systems

Q15 The volume and temperature of air (assumed to be an ideal gas) in a closed vessel is 2.87 m^3 and 300 K , respectively. The gauge pressure indicated by manometer fitted to the wall of the vessel is 0.5 bar . If the gas constant of air is $R = 287 \text{ J/kgK}$ and the atmospheric pressure is 1



bar, the mass of air (in kg) in the vessel is _____
(answer, in integer).

Q16 A gas mixture consists of 3 kg of O_2 , 5 kg of N_2 and 12 kg of CH_4 . The mass fraction and mole fraction of O_2 are

- (A) 0.25 and 0.125 (B) 0.15 and 0.092
(C) 0.25 and 0.092 (D) 0.15 and 0.125

Q17

Zeroth law of thermodynamics is not valid for the following

- (A) 50 ml of water at $25^\circ C$ is mixed with 150 ml of water at $25^\circ C$
(B) 500 ml of milk at $15^\circ C$ is mixed with 100 ml of water at $15^\circ C$
(C) 5 kg of wet steam at $100^\circ C$ is mixed with 50 kg of dry saturated steam at $100^\circ C$
(D) 10 ml of water at $20^\circ C$ is mixed with 10 ml of sulphuric acid at $20^\circ C$



Answer Key

Q1 (B)
Q2 (D)
Q3 (B)
Q4 (B)
Q5 (D)
Q6 212~214
Q7 50~50
Q8 (A)
Q9 (B)

Q10 (A)
Q11 (A)
Q12 (A)
Q13 (A)
Q14 (C)
Q15 5~5
Q16 (B)
Q17 (D)



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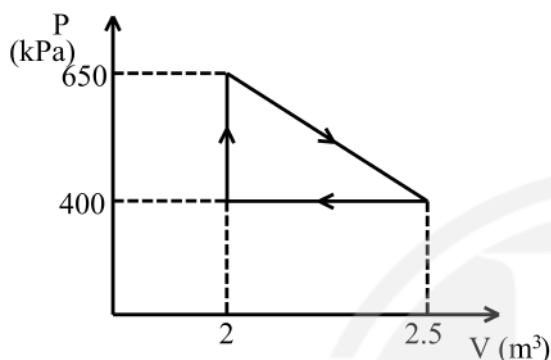
Mechanical Engineering (Hinglish)

Basic Thermodynamics

DPP: 1

Work and Heat

- Q1** An engine operates on the reversible cycle as shown in the figure. The work output from the engine (in kJ/cycle) is _____ (correct to two decimal places).



- Q2** A fluid, contained in a horizontal cylinder fitted with a frictionless leak proof piston, is continuously agitated by means of a stirrer passing through the cylinder cover. The cylinder diameter is 0.40 m. During the stirring process lasting 10 minutes, the piston slowly moves out a distance of 0.485 m against the atmosphere. The net work done by the fluid during the process is 2 kJ. The speed of the electric motor driving the stirrer is 840 rpm. Which one or more of the following statement(s) is/are CORRECT?
- (A) The torque in the shaft is 0.068 N-m to 0.082 N-m.
- (B) The torque in the shaft is 0.058 N-m to 0.062 N-m.
- (C) The power output of the motor is 8.58 W to 9.38 W
- (D) The power output of the motor is 6.58 W to 7.38 W
- Q3** The gas space above water in a closed storage tank contains 'N₂' at 25°C and 100 kPa. Total tank volume is 4m³ and there is 500 kg water at

25°C. An additional 500 kg is now forced into tank. Assuming constant temperature throughout, which one or more of the following statement(s) is/are CORRECT?

- (A) The final pressure of 'N₂' is 114.45 kPa to 118.45 kPa.
- (B) The final pressure of 'N₂' is 110.45 kPa to 112.45 kPa.
- (C) The work done on 'N₂' in the process is 65.20 kJ to 70.20 kJ
- (D) The work done on 'N₂' in the process is 50.20 kJ to 56.20 kJ

- Q4** A gas expands from pressure P_1 to pressure P_2 ($P_2 = P_1/10$). If the process of expansion is isothermal, the volume at the end of expansion is 0.55 m³. If the process of expansion is adiabatic, the volume at the end of expansion will be closer to
- (A) 0.45 m³ (B) 0.55 m³
- (C) 0.65 m³ (D) 0.28 m³

- Q5** A system at a given state undergoes change through the following expansion processes to reach the same final volume:

1. Isothermal
2. Isobaric
3. Adiabatic ($\gamma = 1.4$)
4. Polytropic ($n = 1.3$)

The correct ascending order of the work output in these four processes is

- (A) 1, 2, 4, 3 (B) 1, 4, 3, 2
- (C) 4, 1, 3, 2 (D) 3, 4, 1, 2

- Q6** Consider a refrigerator that consumes 320 W of electric power when it is running. If the refrigerator runs only one quarter of the time and the unit cost of electricity is Rs.0.09/kWh,



the electricity cost of this refrigerator per month (30 days) is

- (A) Rs. 3.56 (B) Rs. 5.18
(C) Rs. 8.54 (D) Rs. 9.28

Q7 Starting from the same initial conditions, an ideal gas expands from volume V_1 to V_2 in three different ways, the work done by the gas is W_1 , if the process is purely isothermal, W_2 if purely isobaric and W_3 if purely adiabatic, then

- (A) $W_2 > W_1 > W_3$
(B) $W_2 > W_3 > W_1$
(C) $W_1 > W_2 > W_3$
(D) $W_1 > W_3 > W_2$

Q8 A system is undergoing a cycle that consists of three processes. During the first process, the work is 5 kJ and the heat is 23 kJ. In the second process no work takes place and the heat interaction is -50 kJ. The third process is adiabatic. The work in the third process is

- (A) -32 kJ (B) -23 kJ
(C) -22 kJ (D) Zero

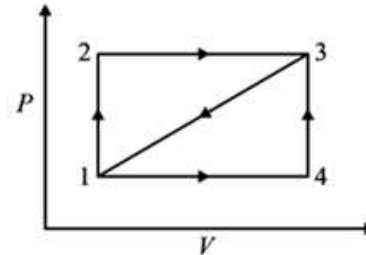
Q9 Air with initial condition of p_1, V_1 expands to final condition of $p_1/2, 3V_1$. The process is
(A) Hyperbolic

(B) Adiabatic

(C) Polytropic with $n > 1$

(D) Polytropic with $n < 1$

Q10 A system absorbs 100 kJ as heat and does 60 kJ work along the path 1-2-3. The same system does 20 kJ work along the path 1-4-3. The heat absorbed during the path 1-4-3 is



- (A) -40 kJ (B) -80 kJ
(C) 80 kJ (D) 60 kJ

Q11 A non-flow quasi-static (reversible) process occurs for which $P = (-3V + 16)$ bar, Where V is volume in m^3 . What is work done when V changes from 2 m^3 to 6 m^3 ?

- (A) $16 \times 10^5 \text{ J}$
(B) $16.5 \times 10^5 \text{ J}$
(C) $16 \times 10^3 \text{ J}$
(D) $16.5 \times 10^2 \text{ J}$



Answer Key

Q1 **58.35~64.35**

Q2 (A, D)

Q3 (A, D)

Q4 (D)

Q5 (D)

Q6 (B)

Q7 (A)

Q8 (A)

Q9 (D)

Q10 (D)

Q11 (A)



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Mechanical Engineering (Hinglish)

Basic Thermodynamics

DPP: 1

First Law of Thermodynamics

Q1 A system executes a cyclic process during which there are two processes as given below:

$${}_1Q_2 = 460 \text{ kJ}, {}_2Q_1 = -100 \text{ kJ} \text{ and } {}_1W_2 = 210 \text{ kJ},$$

What will be work interaction process ${}_2W_1$?

- (A) 100 kJ (B) -210 kJ
(C) 150 kJ (D) -150 kJ

Q2 In the cylinder of an air motor, the compressed air has a specific internal energy of 420 kJ/kg at the beginning of the expansion and a specific internal energy of 200 kJ/kg after expansion. What is the heat flow to or from the cylinder when the work done by the air during the expansion is 10 kJ/kg?

- (A) -120 kJ/kg
(B) -210 kJ/kg
(C) 80 kJ/kg
(D) +100 kJ/kg

Q3 A system is taken through a series of processes as a result of which it is restored to the initial state. The work and heat interactions for some of the processes are measured and they are as given below

Process	W(kJ)	Q(kJ)	U(kJ)
1-2	-100	100	..
2-3	...	-150	200
3-4	-250	...	...
4-1	300	...	50

Network and net heat interaction are (in kJ)

Q4

A stationary mass of gas is compressed without friction from an initial state of 0.3 m^3 and 0.105 MPa to a final state of 0.15 m^3 and 0.105 MPa , the pressure remaining constant during the process. There is a transfer of 40 kJ of heat from the gas during the process. How much does the internal energy of the gas change?

- (A) -24.25 kJ
(B) -19.62 kJ
(C) -15.91 kJ
(D) -12.72 kJ

Q5 In the turbine of a gas turbine unit the gases flow through the turbine at 17 kg/s and the power developed by the turbine is 14000 kW . The specific enthalpies of the gases at inlet and outlet are 1200 kJ/kg and 360 kJ/kg respectively. What is the rate at which heat is rejected from the turbine?

- (A) -89.5 kW
(B) 280 kW
(C) -121.5 kW
(D) 119.3 kW

Q6 A piston-cylinder device contains an ideal gas. The gas undergoes two successive cooling processes by rejecting heat to the surroundings. First the gas is cooled at constant pressure until $T_2 = \frac{3}{4}T_1$. Then the piston is held stationary while the gas is further cooled to $T_3 = \frac{1}{2}T_1$ where all temperatures are in K. Which one or more of the following statement(s) is/are CORRECT?

- (A) The ratio of the final volume to the initial volume of the gas $3/4$
(B) The ratio of the final volume to the initial volume of the gas $2/3$



(C) The work done on the gas by the piston is $\frac{RT_1}{4}$

(D) The total heat transferred from the gas is $(c_p + c_v) \frac{T_1}{4}$

- Q7** A slow – chemical reaction takes place in a fluid at the constant pressure of 0.1 MPa. The fluid is surrounded by a perfect heat insulator during the reaction which begins at state 1 and ends at state 2. The insulation is then removed and 105 kJ if heat flows to the surroundings as the fluid goes to state 3. The following data are observed for the fluid at states 1, 2, 3.

State	1	2	3
Volume (m^3)	0.03	0.03	0.06

For the fluid system, calculate E_2 and E_3 , if $E_1 = 0$.

- (A) 29.7 kJ & 110.7 kJ
 (B) – 29.7 kJ & –110.7 kJ
 (C) 29.7 kJ & zero
 (D) zero & 110.7 kJ
- Q8** The heat capacity at constant pressure of a certain system is a function of temperature only and may be expressed as
- $$C_P = 0.239 + \frac{41.87}{t+100} \text{ J/}^\circ\text{C}$$
- Where t is the temp of the system in $^\circ\text{C}$. The system is heated while it is maintained at a pressure of 1 atm until its volume increases from 2000 cm^3 to 2400 cm^3 and its temp increases from 0°C to 100°C .
- Find the magnitude of heat interaction
- (A) 238.32 J (B) 219.24 J
 (C) 238.32 kJ (D) 219.22 kJ
- Q9** The heat capacity at constant pressure of a certain system is a function of temperature only and may be expressed as
- $$C_P = 0.239 + \frac{41.87}{t+100} \text{ J/}^\circ\text{C}$$

Where t is the temp of the system in $^\circ\text{C}$. The system is heated while it is maintained at a pressure of 1 atm until its volume increases from 2000 cm^3 to 2400 cm^3 and its temp increases from 0°C to 100°C .

How much does the internal energy increase?

- (A) 197.79 kJ (B) 197.79 J
 (C) 179.24 kJ (D) 179.24 J
- Q10** A 2-kW baseboard electric resistance heater in a vacant room is turned on and kept on for 15 min. The mass of air in the room is 75 kg, and the room is tightly sealed so that no air can leak in or out. The temperature rise of air at the end of 15 min is
- (A) 8.5°C (B) 12.4°C
 (C) 24.0°C (D) 33.4°C
- Q11** In a test of a water-jacketed compressor, the shaft work required is 90 kN-m/kg of air compressed. During compression, increase in enthalpy of air is 30 kJ/kg of air and increase in enthalpy of circulating cooling water is 40 kJ/kg of air. The change in velocity is negligible. The amount of heat lost to the atmosphere from the compressor per kg of air is
- (A) 20 kJ (B) 60 kJ
 (C) 80 kJ (D) 120 kJ
- Q12** A room for the four persons has two fans, each consuming 0.18 kW power, and three 100 W lamps. Ventilation air at the rate of 80 kg/h enters with an enthalpy of 84 kJ/kg and leaves with an enthalpy of 59 kJ/kg. If each person puts out heat at the rate of 630 kJ/h, determine the rate at which heat is to be removed by a room cooler, so that a steady state is maintained in the room.
- (A) –1.91 kW
 (B) 1.91 kW
 (C) 0.59 kW
 (D) – 3.59 kW
- Q13** An ideal-gas mixture of helium and argon gases with identical mass fractions enters a



turbine at 1200 K and 1 MPa at a rate 0.3 kg/s, and expands reversible adiabatic to 100 kPa.

The power output of the turbine is:

- (A) 478 kW (B) 619 kW
(C) 926 kW (D) 729 kW

- Q14** A pressure cylinder of volume V contains air at pressure P_1 and temperature T_1 . It is to be filled from a compressed air line maintained at constant pressure P_1 and temperature T_1 . The temperature of the air in the cylinder after it has been charged to the pressure of the line is

- (A) $T_2 = \frac{\gamma T_1}{1 + \frac{P_1}{P_i} \left(\gamma \frac{T_i}{T_1} - 1 \right)}$
 (B) $T_2 = \frac{\gamma T_1}{1 - \frac{P_1}{P_i} \left(\gamma \frac{T_i}{T_1} - 1 \right)}$
 (C) $T_2 = \frac{T_1}{1 + \frac{P_1}{P_i} \left(\gamma \frac{T_i}{T_1} - 1 \right)}$
 (D) $T_2 = \frac{\gamma T_1}{1 + \frac{P_1}{P_i} \left(\gamma \frac{T_i}{T_1} + 1 \right)}$



Answer Key

Q1 (C)

Q2 (B)

Q3 -200~-200

Q4 (A)

Q5 (B)

Q6 (A, C, D)

Q7 (B)

Q8 (A)

Q9 (B)

Q10 (D)

Q11 (A)

Q12 (B)

Q13 (B)

Q14 (A)

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Basic Thermodynamics

DPP: 1

Second Law of Thermodynamics

Q1 A refrigerator operates on Reversed Carnot cycle. What is the power required to drive the refrigerator between temperatures of 42°C and 4°C , if heat at the rate of 2 kJ/s is extracted from the low temperature region?

- (A) 0.174 kW (B) 0.374 kW
(C) 0.274 kW (D) 0.474 kW

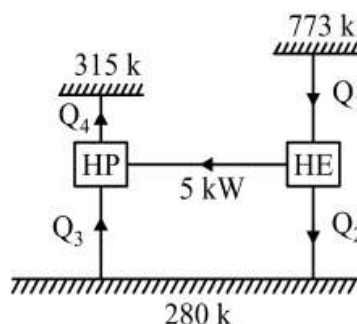
Q2 A solar energy-based heat engine which receives 80 kJ of heat at 100°C and rejected 70 kJ of heat to the ambient at 30°C is to be designed. The thermal efficiency of the heat engine is ____

- (A) 70%
(B) 1.88%
(C) 12.5%
(D) None of the above

Q3 Two Carnot heat engines are operating in series such that the heat sink of the first engine serves as the heat source of the second one. If the source temperature of the first engine is 1327°C , the sink temperature of the second engine is 127°C , and the thermal efficiency of both the engines are the same, the temperature of the intermediate reservoir is:

- (A) 627°C (B) 410°C
(C) 577°C (D) 527°C

Q4 A Carnot engine (HE) developing 5 kW drives a heat pump (HP) is shown in figure. Determine the value of Q_4 and Q_2 (in kW) are, respectively.



- (A) 7.84 & 45 (B) 45 & 2.84
(C) 2.84 & 45 (D) 45 & 7.84

Q5 A domestic food freezer maintains a temperature of -15°C . The ambient air temperature is 30°C . If heat leaks into the freezer at the continuous rate of 1.75 kJ/s , the least power necessary to pump this heat out continuously will be nearly

- (A) 0.1 kW (B) 0.2 kW
(C) 0.3 kW (D) 0.4 kW

Q6 Consider the following statements:

1. The entropy of a pure crystalline substance at absolute zero temperature is zero.
2. The efficiency of a reversible heat engine is independent of the nature of the working substance and depends only on the temperature of the reservoirs between which it operates.
3. Carnot's theorem states that of all heat engines operating between a given constant temperature source and a given constant temperature sink, none has a higher efficiency than a reversible engine.

Which of the above statements are correct?

- (A) 1 and 2 only (B) 1 and 3 only
(C) 2 and 3 only (D) 1, 2 and 3



- Q7** Two reversible engines are connected in series between a heat source and a sink. The efficiencies of these engines are 60% and 50% respectively. If these two engines are replaced by a single reversible engine, the efficiency of this engine will be
 (A) 60% (B) 70%
 (C) 80% (D) 90%
- Q8** A heat pump with a COP of 3.2 is used to heat a perfectly sealed house (no leaks). The entire mass within the house (air, furniture, etc.) is equivalent to 1200 kg of air. When running the heat pump consumes electric power at a rate of 5 kW. The temperature of the house was 7°C when the heat pump was turned on. If heat transfer through the envelope of the house (walls, roof, etc.) is negligible, the length of time the heat pump must run to raise the temperature of the entire contents of the house to 22°C is:
 (A) 13.5 min (B) 138 min
 (C) 43.1 min (D) 188 min
- Q9** A typical new household refrigerator consumes about 680 kWh of electricity per year and has a coefficient of performance of 1.4. The amount of heat removed by this refrigerator from the refrigerated space per year is
 (A) 952 MJ/yr (B) 1749 MJ/yr
 (C) 2446 MJ/yr (D) 3427 MJ/yr
- Q10** Using the engine of 30% thermal efficiency to drive a refrigerator having a COP of 5, what is the heat input into the engine for each MJ removed from the cold body by the refrigerator? _____ MJ (round off to two decimal places).
- Q11** A heat pump is used in place of a furnace for heating a house in winter. The heat lost from the house is 60,000 kJ/hr when the outside air temperature is -10°C and the inside is maintained at 21°C. Determine the minimum electric power required to operate the heat pump. Compare it with the power required to maintain the room at 21°C with an ordinary electric resistance heater:
- Q12** A house is to be maintained at a temperature of 20°C by means of a heat pump pumping heat from the atmosphere. Heat losses through the walls of the house are estimated at 0.65 kW per unit of temperature difference between the inside of the house and the atmosphere. Which one or more of the following statement(s) is/are CORRECT?
- (A) If the atmospheric temperature is -10°C, the minimum power required to drive the pump is 1.9 kW to 2.2 kW.
 (B) If the atmospheric temperature is -10°C, the minimum power required to drive the pump is 0.9 kW to 1.2 kW.
 (C) It is proposed to use the same heat pump to cool the house in summer. For the same room temperature, the same heat loss rate, and the same power input to the pump, the maximum permissible atmospheric temperature is 48 °C to 52 °C.
 (D) It is proposed to use the same heat pump to cool the house in summer. For the same room temperature, the same heat loss rate, and the same power input to the pump, the maximum permissible atmospheric temperature is 58 °C to 62 °C.



Answer Key

Q1 (C)
Q2 (C)
Q3 (D)
Q4 (B)
Q5 (C)
Q6 (D)

Q7 (C)
Q8 (A)
Q9 (D)
Q10 0.65~0.69
Q11 6321.5~6329.5
Q12 (A, C)



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Mechanical Engineering [Hinglish]

Basic Thermodynamics

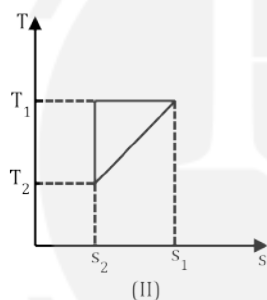
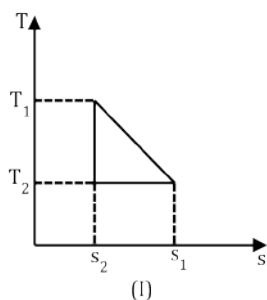
DPP: 1

Entropy

Q1 In which one of the following situations the entropy change will be negative?

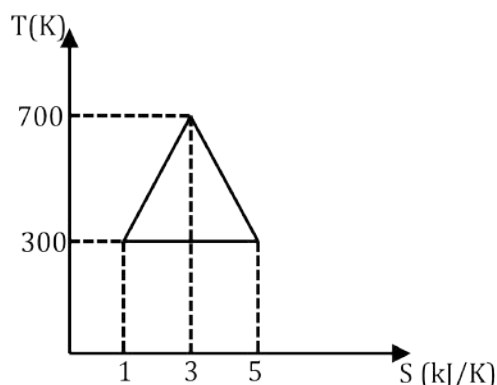
- (A) Air expands isothermally from 6 bars to 3 bar
- (B) Air is compressed to half the volume at constant pressure
- (C) Heat is supplied to air at constant volume till the pressure becomes three folds
- (D) Air expands isentropically from 6 bars to 3 bar

Q2 What is the ratio of the efficiencies $\left(\frac{\eta_I}{\eta_{II}}\right)$ for the two cycles as shown in the T-s diagram?



- (A) $\frac{T_1+T_2}{2T_1}$
- (B) $\frac{T_1+T_2}{2T_2}$
- (C) $\frac{2T_2}{T_1+T_2}$
- (D) $\frac{2T_1}{T_1+T_2}$

Q3 The thermal efficiency of the hypothetical cycle shown is



- (A) 0.6
- (B) 0.5
- (C) 0.4
- (D) 0.04

Q4 A liquid of heat capacity 5 J/K in an insulated container is heated electrically from 300 K to 600 K. If $\ln 2 = 0.693$, entropy generation of the universe would be

- (A) 6.93 J/K
- (B) 3.465 J/K
- (C) 34.65 J/K
- (D) 10.65 J/K

Q5 Match the curves in Diagram I with the curves in Diagram II and select the correct answer.

Diagram I (Process on p-v plane)

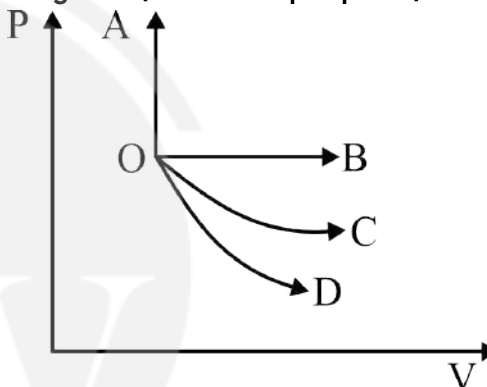
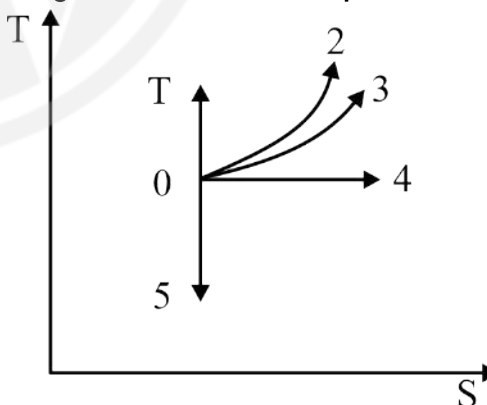


Diagram II (Process on T-s plane)



- (A) A-3, B-2, C-4, D-5
- (B) A-2, B-3, C-4, D-5
- (C) A-2, B-3, C-4, D-1
- (D) A-1, B-4, C-2, D-3

Q6 A cyclic heat engine receives 600 kJ of heat from a 1000 K source and rejects 450 kJ to a



300 K sink. The quantity $\oint \frac{\delta Q}{T}$ and efficiency of the engine are respectively.

- (A) 2.1 kJ/K and 70%
- (B) - 0.9 kJ/K and 25%
- (C) +0.9 kJ/K and 70%
- (D) - 2.1 kJ/K and 25%

Q7 A heat engine receives 1000 kW of heat at a constant temperature of 285°C and rejects 492 kW of heat at 50°C. Consider the following thermodynamic cycles in this regard:

1. Carnot cycle
2. Reversible cycle
2. Irreversible cycle

Which of these cycles could possibly be executed by the engine?

- (A) 1 alone
- (B) 3 alone
- (C) 1 and 2
- (D) None of 1, 2 and 3

Q8 Steam is condensed at a constant temperature of 30°C as it flows through the condenser of a power plant by rejecting heat at a rate of 55 MW. The rate of entropy change of steam as it flows through the condenser is

- (A) -1.83 MW/K
- (B) -0.18 MW/K
- (C) 0.56 MW/K
- (D) 0 MW/K

Q9 A unit mass of an ideal gas at temperature T undergoes a reversible isothermal process from pressure P_1 to pressure P_2 while losing heat to the surroundings at temperature T in the amount of q . If the gas constant of the gas is R , the entropy change of the gas Δs during this process is

- (A) $\Delta s = R \ln \left(\frac{P_2}{P_1} \right)$
- (B) $\Delta s = R \ln \left(\frac{P_2}{P_1} \right) - q/T$
- (C) $\Delta s = R \ln (P_1/P_2)$
- (D) $\Delta s = 0$

Q10

Air is compressed from room conditions to a specified pressure in a reversible manner by two compressors one isothermal and the other adiabatic. If the entropy change of air Δ_{sisot} during the reversible isothermal compression, and Δ_{sadia} during the reversible adiabatic compression, the correct statement regarding entropy change of air per unit mass is

- (A) $\Delta_{sisot} = \Delta_{sadia} = 0$
- (B) $\Delta_{sisot} = \Delta_{sadia} > 0$
- (C) $\Delta_{sadia} = 0$
- (D) $\Delta_{sisot} < 0$

Q11 In a Carnot cycle, heat is supplied at 350°C and rejected at 27°C. The working fluid is water which, while receiving heat, evaporates from liquid at 350°C to steam at 350°C. The associated entropy change is 1.44 kJ/kg K. Which one or more of the following statement(s) is/are CORRECT?

- (A) If the cycle operates on a stationary mass of 1 kg of water, the work done per cycle is 460.5 kJ/kg to 470.5 kJ/kg.
- (B) If the cycle operates on a stationary mass of 1 kg of water, the heat supplied per cycle is 893.5 kJ/kg to 901.7 kJ/kg
- (C) If the cycle operates in steady flow with a power output of 20 kW, the steam flow rate is 0.031 kg/s to 0.035 kg/s.
- (D) If the cycle operates in steady flow with a power output of 20 kW, the steam flow rate is 0.041 kg/s to 0.045 kg/s.

Q12 Water is heated at a constant pressure of 0.7 MPa. The boiling point is 164.97°C. The initial temperature of water is 0°C. The latent heat of evaporation is 2066.3 kJ/kg. Find the increase of entropy of water (kJ/kg-K), if the final state is steam.

Q13 A Carnot cycle operates between the temperature limits of 300 K and 2000 K, and produces 600 kW of net power. The rate of



entropy change of the working fluid during the heat addition process is

(A) 0

(B) 0.300 kW/K

(C) 0.353 kW/K

(D) 0.261 kW/K



Answer Key

Q1 (B)
Q2 (D)
Q3 (C)
Q4 (B)
Q5 (B)
Q6 (B)
Q7 (D)

Q8 (B)
Q9 (C)
Q10 (D)
Q11 (A, B, D)
Q12 6.6~6.8
Q13 (C)



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Mechanical Engineering [Hinglish]

Basic Thermodynamics

DPP: 1

Exergy

- Q1** Calculate the specific exergy of air for a state at 2 bar, 393.15 K when the surroundings are at 1 bar, 293.15 K. Take $C_p = 1$ and $R = 0.287$ kJ/kg K (answer in one decimal place).
- Q2** Air expands in a turbine adiabatically from 500 kPa, 400 K and 150 m/s to 100 kPa, 300 K and 70 m/s. The environment is at 100 kPa, 17°C. Which one or more of the following statements is/are CORRECT?
 (A) The maximum work output (kJ/kg) is 159.35
 (B) The actual work output (kJ/kg) is 108.8
 (C) The actual work output (kJ/kg) is 118.8
 (D) The irreversibility (kJ/kg) is 50.55
- Q3** A source at a temperature of 500 K provides 1000 kJ of heat. The temperature of environment is 27°C. The unavailable energy out of this heat energy (in kJ) is ____ (round off to nearest integer).
- Q4** A heat engine receives half of its heat supply at 1000 K and half at 500 K while rejecting heat to a sink at 300 K. The maximum thermal efficiency of the heat engine is
 (A) 55% (B) 50%
 (C) 52% (D) 58%
- Q5** A heat engine that receives heat from a furnace at 1200°C and rejects waste heat to a river at 20°C has a thermal efficiency of 40%. Determine second law efficiency of power plant is
 (A) 74.6% (B) 49.9%
 (C) 35.6% (D) 52.6%
- Q6**
- Argon gas expands in an adiabatic turbine from 3 MPa and 750°C to 0.2 MPa at a rate of 5 kg/s. The maximum power output of the turbine is
 (A) 1.06 MW (B) 1.29 MW
 (C) 1.43 MW (D) 1.76 MW
- Q7** Heat is lost through plane walls steady at a rate of 800W. If the inner and outer surface temperatures of the wall are 20°C and 5°C respectively and environment temperature is 0°C the rate of energy destroyed within wall is
 (A) 40 W (B) 17,500 W
 (C) 765 W (D) 32,800 W
- Q8** A house that is losing heat at a rate of 50,000 kJ/hr when outside temperature drops to 4°C is to be heated by electric resistance heaters. If house is to be maintain at 25°C at all times, the irreversibility (in kW) is ____ (round off to one decimal place).
- Q9** A pump raises pressure of saturated liquid water at 100 kPa (density $\rho = 959$ kg/m³) to 2 MPa. The isentropic efficiency of the pump is 0.92. The work done by the pump (in J/kg) is _____. (round off to nearest integer).
- Q10** An ideal gas of 1 kg mass enclosed inside a rigid vessel at the initial temperature 1200 K is employed as a heat source. The specific heat C_v of the gas is 718 J/kgK. The maximum work in kJ that can be developed by operating a heat engine between the ideal gas and the ambient at 300K is
 (A) 646.2 (B) 484.7
 (C) 387.7 (D) 347.6



Answer Key

Q1 70.5~74.5

Q2 (A, B, D)

Q3 600~600

Q4 (A)

Q5 (B)

Q6 (D)

Q7 (A)

Q8 12.7~13.1

Q9 2152~2154

Q10 (D)



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Mechanical Engineering (Hinglish)

Basic Thermodynamics

DPP: 1

Properties of Pure Substance

- Q1** Steam expands in a turbine from 4 MPa and 500°C to 0.5 MPa and 250°C at a rate of 1350 kg/h. Heat is lost from the turbine at a rate of 25 kJ/s during the process. The power output of the turbine is
 (A) 157 kW (B) 207 kW
 (C) 182 kW (D) 287 kW
- Q2** 3-m³ rigid vessel contains steam at 10 MPa and 500°C. The mass of the steam is
 (A) 3.0 kg (B) 19 kg
 (C) 84 kg (D) 91 kg
- Q3** A 300-m³ rigid tank is filled with saturated liquid vapor mixture of water at 200 kPa. If 25 percent of the mass is liquid and 75 percent of the mass is vapor, the total mass in the tank is
 (A) 451 kg (B) 556 kg
 (C) 300 kg (D) 331 kg
- Q4** Steam enters an adiabatic turbine at 8 MPa and 500°C at a rate of 18 kg/s, and exits at 0.2 MPa and 300°C. The rate of entropy generation in the turbine is
 (A) 0 kW/K (B) 7.2 kW/K
 (C) 21 kW/K (D) 15 kW/K
- Q5** Steam expands in an adiabatic turbine from 8 MPa and 500°C to 0.1 MPa at a rate of 3 kg/sec. If steam leaves the turbine as saturated vapor, the power output of the turbine is
 (A) 2174 kW (B) 698 kW
 (C) 288 kW (D) 1674 kW
- Q6** A piston-cylinder device contains 5 kg of saturated water vapor at 3 MPa. Now heat is rejected from the cylinder at constant pressure until the water vapor completely condenses, so that the cylinder contains saturated liquid at 3 MPa at the end of the process. The entropy change of the system during this process is
 (A) 0 kJ/K
 (B) -3.5 kJ/K
 (C) -12.5 kJ/K
 (D) -17.7 kJ/K
- Q7** A heat engine cycle is executed with steam in the saturation dome between the pressure limits of 5 and 2 MPa. If heat is supplied to the heat engine at a rate of 380 kJ/s, the maximum power output of this heat engine is
 (A) 36.5 kW (B) 74.2 kW
 (C) 186.2 kW (D) 343.5 kW
- Q8** Water is boiled at 1 atm pressure in a coffee maker equipped with an immersion-type electric heating element. The coffee maker initially contains 1 kg of water. Once boiling started, it is observed that half of the water in the coffee maker evaporated in 18 minutes. If the heat loss from the coffee maker is negligible, the power rating of the heating element is
 (A) 0.90 kW (B) 1.52 kW
 (C) 2.09 kW (D) 1.05 kW
- Q9** A room is filled with saturated steam at 100°C. Now a 5-kg bowling ball at 25°C is brought to the room. Heat is transferred to the ball from the steam, and the temperature of the ball rises to 100°C while some steam condenses on the ball as it loses heat (but it still remains at 100°C). The specific heat of the ball can be taken to be 1.8 kJ/kg·°C. The mass of steam that condensed during this process is
 (A) 80 g (B) 128 g
 (C) 299 g (D) 351 g



- Q10** Boiler steam at 8 bar, 250°C , reaches the engine control valve through a pipeline at 7 bar, 200°C . It is throttled to 5 bar before expanding in the engine to 0.1 bar, 0.9 dry. Determine per kg of steam
- (A) the heat loss in the pipe line is 105.4 kJ/kg .

- (B) the temperature drop in passing through the throttle valve is 4.8°C
- (C) the work output of the engine is 396.2 kJ/kg .
- (D) the entropy change due to throttling is 0.1439 kJ/kg-K



Answer Key

Q1 (A)
Q2 (D)
Q3 (A)
Q4 (C)
Q5 (A)

Q6 (D)
Q7 (A)
Q8 (D)
Q9 (C)
Q10 (A, B, C, D)



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Mechanical Engineering (Hinglish)

Basic Thermodynamics

DPP: 1

Thermodynamic Relations

- Q1** For an ideal gas, $\left(\frac{\partial T}{\partial v}\right)_s$ is equal to
 (A) $\frac{R}{P}$ (B) $-\frac{P}{c_v}$
 (C) $\frac{v}{c_p}$ (D) $-\frac{T}{c_v}$
- Q2** For a gas, the equation of state is given by $P(v - b) = RT$, then using Maxwell relations, $\left(\frac{\partial s}{\partial P}\right)_T$ is equal to
 (A) $\frac{R}{v}$ (B) RT
 (C) $-\frac{R}{P}$ (D) $\frac{R}{P}$
- Q3** For a substance, the equation of state is given by $P(v - a) = RT$. The expression for the volume expansivity (α) is _____.
 (A) $\frac{R}{T + aP}$ (B) $\frac{R}{RT + aP}$
 (C) $\frac{R}{T - av}$ (D) $\frac{R}{RT - aP}$
- Q4** Liquid methanol at 25°C is compressed isentropically in a insulated piston cylinder arrangement from 100 kPa to 10 MPa. At 25°C , the liquid methanol has an isentropic compressibility of $1.05 \times 10^{-9} \text{ m}^2/\text{N}$. The specific work required to compress the liquid methanol is (in J/kg) _____. Take density of liquid methanol is 787 kg/m^3 .
 (A) 52.7 (B) 70.2
 (C) 60.7 (D) 66.7
- Q5** Which gas shows a heating effect in the Joule-Thomson experiment, while undergoing throttling process through a porous plug of cotton wool?
 (A) Carbon dioxide (B) Hydrogen
 (C) Oxygen (D) Nitrogen
- Q6** Joule-Thomson coefficient is the slope of
 (A) constant enthalpy lines on $T - s$ diagram
 (B) constant enthalpy lines on $T - p$ diagram
 (C) inversion curve on $T - s$ diagram
 (D) inversion curve on $T - p$ diagram
- Q7** Consider the following statement pertaining to Clapeyron equation:
 1. It is useful in estimating properties like enthalpy from other measurable properties.
 2. At a change of phase, it can be used to find the latent heat at a given pressure.
 3. It is derived from the relationship $\left(\frac{\partial p}{\partial v}\right)_T = \left(\frac{\partial s}{\partial T}\right)_v$
 Which of the above statements are correct?
 (A) 1 and 3 only (B) 2 and 3 only
 (C) 1 and 2 only (D) 1, 2 and 3
- Q8** Which of the following relationships represents the change of entropy of a perfect gas?
 1. $C_p \frac{dT}{T} + \frac{R}{V} dV$
 2. $C_p \frac{dT}{T} - \frac{R}{P} dP$
 3. $C_v \frac{dP}{P} + C_p \frac{dV}{V}$
 4. $C_v \frac{dP}{P} - C_v \frac{dV}{V}$
 (A) 1, 2 and 4 only
 (B) 1 and 3 only
 (C) 2 and 3 only
 (D) 1, 2 and 4
- Q9** Joule-Thomson coefficient for an ideal gas is
 (A) Higher than zero
 (B) Higher than zero
 (C) zero
 (D) 1
- Q10** A positive value of Joule-Thomson coefficient of a fluid means
 (A) Temperature drops during throttling
 (B) Temperature remains constant during throttling
 (C) Temperature rises during throttling
 (D) None of these



Answer Key

Q1 (B)

Q2 (C)

Q3 (B)

Q4 (D)

Q5 (B)

Q6 (B)

Q7 (C)

Q8 (C)

Q9 (C)

Q10 (A)



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